The Friendly Guide to

What's Happening Inside an AI's Brain

(explained so a 5-year-old gets it, but an expert learns something too)

Mechanistic Interpretability with Sparse Autoencoders

CS 463 | University of San Francisco | Spring 2026

math code facts creative emotions reasoning

Table of Contents

Our adventure through the AI brain, chapter by chapter

Chapter 1

What Is an AI?

The Word-Guessing Robot

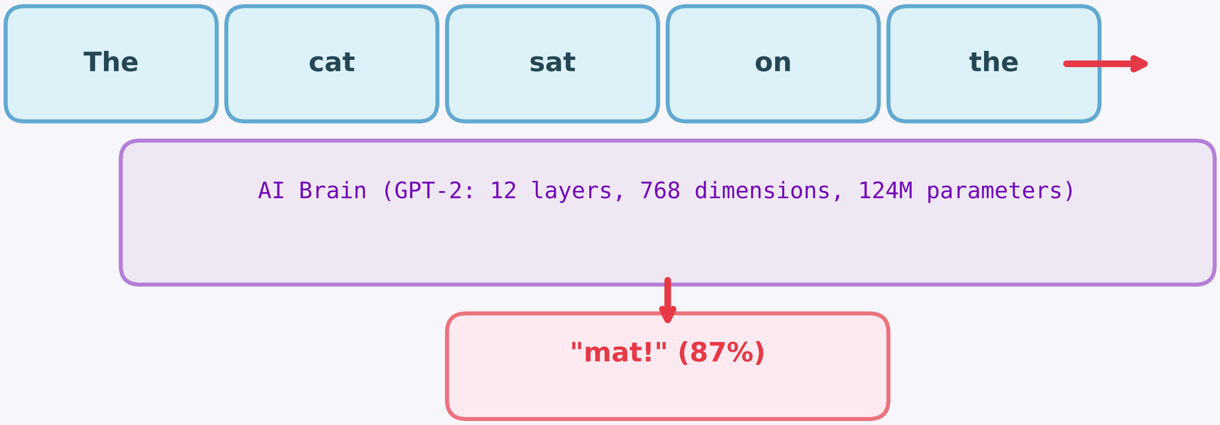
If I were 5...

Imagine you have a robot friend who is REALLY good at one game: you say the beginning of a sentence, and the robot guesses what word comes next. You say "The cat sat on the..." and the robot says "mat!" That's basically what an AI language model does. It reads your words and guesses the next one, over and over, until it writes a whole answer.

The real deal:

A Large Language Model (LLM) like GPT-2 is an autoregressive transformer that predicts P(next_token | previous_tokens). It has 124M parameters organized in 12 layers, each containing multi-head attention (12 heads) and an MLP. The model processes text as tokens (subwords) through a 768-dimensional residual stream.

How the Word-Guessing Game Works



Chapter 2

Layers

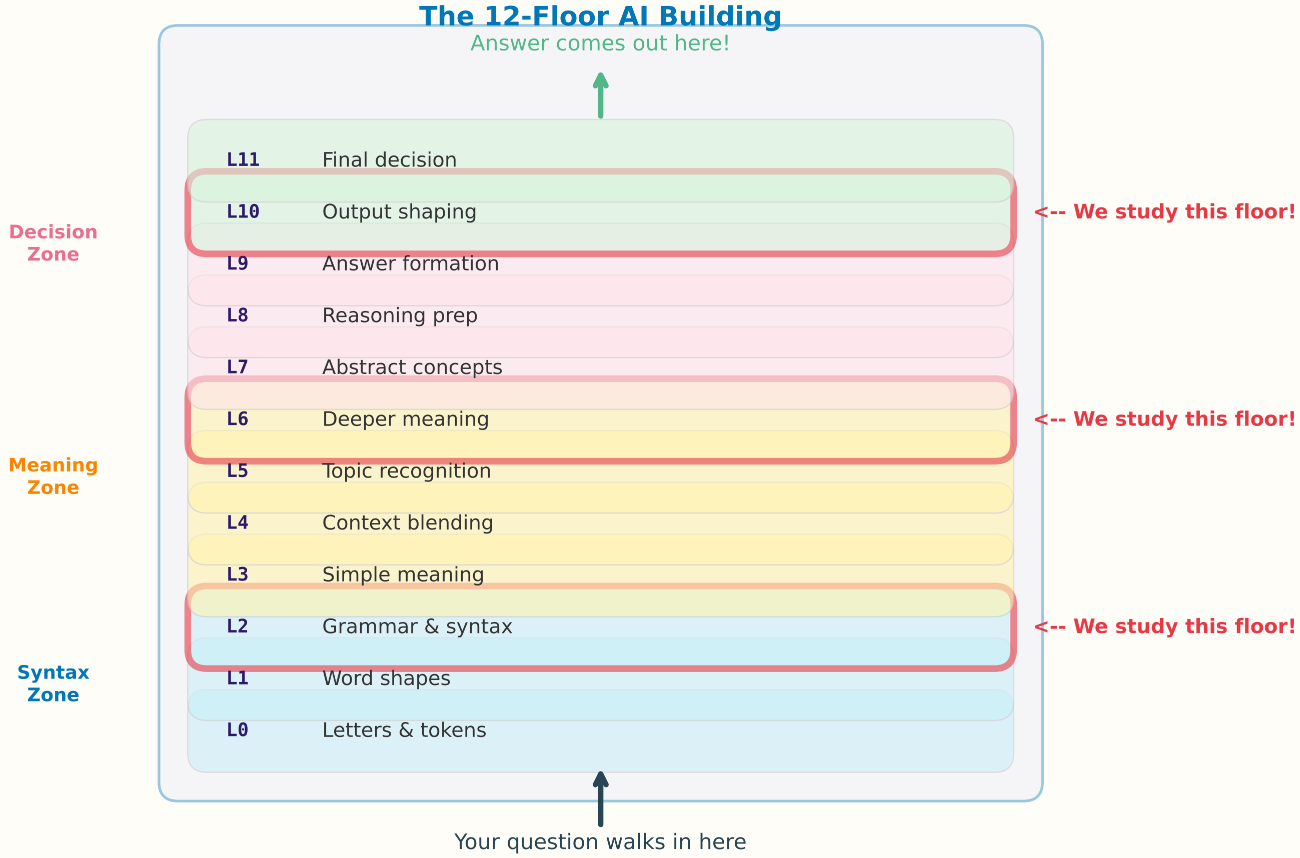
Floors in a Tall Building

If I were 5...

The robot's brain is like a tall building with 12 floors. When you ask a question, your words walk in through the front door (floor 1) and climb up, floor by floor. On each floor, the words get changed a little bit -- they pick up new ideas. By the time they reach the top floor (floor 12), they've become the answer! The bottom floors figure out basic stuff ("this is English"), the middle floors figure out meaning ("this is a math question"), and the top floors decide what to say.

The real deal:

Each of GPT-2's 12 layers consists of multi-head self-attention followed by an MLP, with residual connections. The residual stream is the central 768-dim vector that accumulates information layer by layer. Early layers (0-3) handle syntax and token identity; middle layers (4-8) develop semantic representations; late layers (9-11) prepare the final next-token distribution via the unembedding matrix.



Chapter 3

Superposition

Too Many Toys in One Box

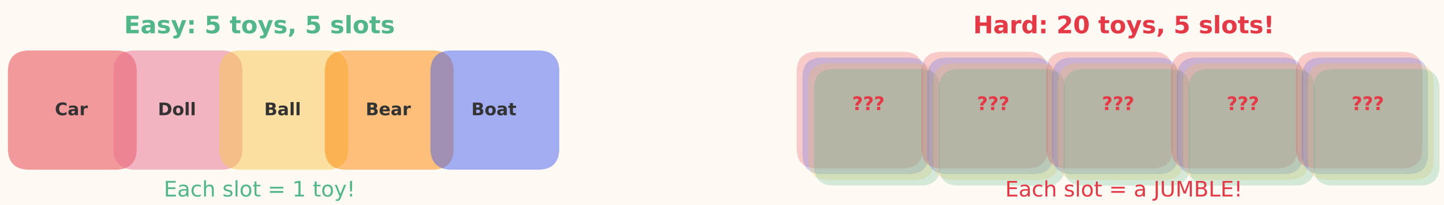
If I were 5...

Imagine you have a toy box with only 5 slots, but you need to store 100 different toys. What do you do? You get clever! You stack the teddy bear AND the puzzle piece in the same slot. You squeeze the race car AND the doll together. Each slot now holds a MIXTURE of toys. If someone asks "what's in slot 3?" you can't give a simple answer -- it's a jumble. That's superposition: the AI's brain has 768 slots but needs to remember THOUSANDS of ideas, so it stuffs multiple ideas into each slot.

`The real deal:`

Superposition occurs when a neural network represents more features (concepts) than it has dimensions. GPT-2 has d_model=768, but encodes thousands of distinct concepts. Features are stored as nearly-orthogonal directions in a high-dimensional space. Individual neurons become polysemantic -- each neuron responds to multiple unrelated concepts. This makes naive per-neuron interpretation unreliable. (Elhage et al., 'Toy Models of Superposition', 2022)

The Toy Box Problem



This is what happens in GPT-2's brain:

`768 neurons` (the slots)	must hold	`1000s of concepts` (the toys)

Chapter 4

Sparse Autoencoders

The Magic Sorting Machine

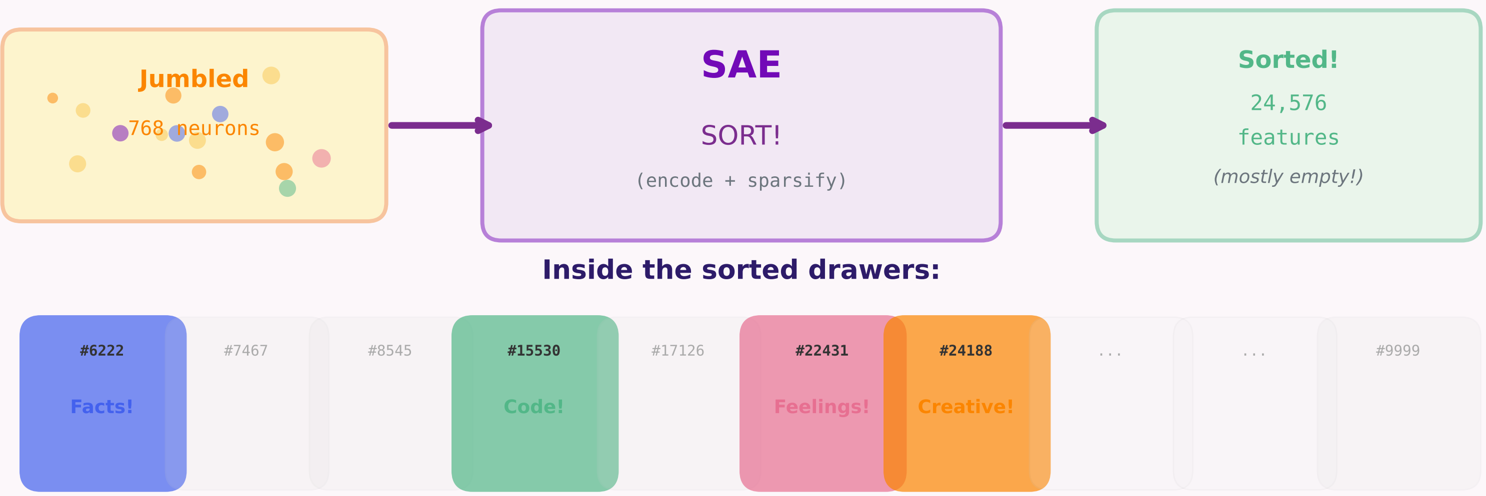
If I were 5...

Remember the messy toy box where everything was jumbled together? Now imagine you have a MAGIC SORTING MACHINE. You dump in the jumbled mess, and out comes 24,576 tiny labeled drawers -- and each drawer has EXACTLY ONE toy in it! Drawer #15530 has the "code toy", drawer #6222 has the "facts toy", and so on. Most drawers are empty (the machine only opens the ones that matter for your question). This magic sorter is called a Sparse Autoencoder!

The real deal:

An SAE is trained to reconstruct the 768-dim residual stream activation through a bottleneck: encode(x) = ReLU(W_enc @ x + b_enc) producing a 24,576-dim sparse vector, then decode(z) = W_dec @ z + b_dec reconstructing the original 768 dims. The sparsity penalty (L1) ensures most of the 24,576 features are zero for any given input -- typically only 50-300 are nonzero. Each feature ideally corresponds to one monosemantic concept. We use the gpt2-small-res-jb release by Joseph Bloom (24,576 features per layer).

The Magic Sorting Machine (SAE)



Most drawers are EMPTY -- only the relevant ones light up!

(This is what "sparse" means -- very few are active at once.)

Chapter 5

Features

The Individual LEGO Bricks

If I were 5...

You know how a LEGO castle is built from hundreds of tiny bricks? Each brick is simple -- a red 2x4, a blue 1x2, a green flat piece. But together they make something complex. Features are like those bricks! Feature #6222 might mean "the AI is thinking about a fact." Feature #15530 might mean "the AI is thinking about computer code." Each feature is one simple idea. When you ask the AI a question, different bricks (features) light up depending on what kind of question it is!

The real deal:

Each of the 24,576 SAE features is a learned direction in the residual stream space. When a feature 'fires' (nonzero activation), it means the input triggered that particular concept. Features are monosemantic by design -- the L1 penalty during SAE training encourages each feature to respond to one thing. Our analysis identifies discriminative features: those that fire selectively for one prompt category. E.g., feature #6222 has a selectivity score of 2.11 for reasoning prompts -- the strongest single-category signal in our study. Feature #7484 scores 1.56 for factual, #1881 scores 1.17 for code.

What Features Look Like (Layer 6 -- the middle floor)

Feature #22431 "*Thinking Brick*" 2.1

Fires for complex reasoning & emotions.
"If all birds can fly, and penguins..."

Feature #7484 "*Facts Brick*" 1.6

Fires when AI thinks about facts.
"What is the capital of France?"

Feature #1881 "*Code Brick*" 1.2

Fires when AI thinks about code.
"Write a function to sort a list"

Feature #13534 "*Creativity Brick*" 1.1

Fires when AI thinks creatively.
"Write a haiku about autumn"

The longer the bar, the MORE selective that feature is for its category.

Chapter 6

Our Experiment

Asking 300 Real Human Questions

If I were 5...

We played a big game of "20 Questions" -- but instead of 20, we asked the robot 300 questions from REAL people! We found them in two big collections of questions that real humans asked to AI chatbots (Search Arena and WildChat). We split them into 6 types: math problems, coding puzzles, trivia facts, creative writing prompts, emotional feelings, and brain teasers -- 50 of each! Then we peeked inside the robot's brain to see which drawers lit up.

The real deal:

We sourced 300 real human prompts (50 per category) from Search Arena 24k and WildChat 4.8M. For each prompt, we ran the model forward, extracted the last-token residual stream at layers 2, 6, and 10, and passed it through the corresponding SAE to get a 24,576-dim feature activation vector. Total: 300 prompts x 3 layers = 900 feature vectors.

The 6 Types of Questions We Asked

1 **MATH**
"What is 7 x 8?"
50 questions

2 **CODE**
"Sort this list"
50 questions

3 **FACTS**
"Capital of France?"
50 questions

4 **CREATIVE**
"Write a haiku"
50 questions

5 **EMOTIONAL**
"I feel overwhelmed"
50 questions

6 **REASONING**
"If all dogs are pets..."
50 questions

Total: 300 prompts --> 900 brain scans (3 floors each!)

Chapter 7

Real Questions Beat Fake Ones

Why We Threw Out Our First Dataset

If I were 5...

When we started, we MADE UP our 120 questions ourselves -- like "What is 7 plus 8?" and "Write a poem about a cat." We thought we were being clever! But our teacher said: "Hey, those are YOUR ideas of what people ask. Real people are MESSIER!" Real people type weird stuff, mix languages, ramble, use slang. So we threw out our 120 fake questions and got 300 REAL ones from actual people on the internet. The robot meets the real world!

The real deal:

Per instructor feedback at Milestone 1, hand-curated synthetic prompts risk reflecting the researcher's intuition about category boundaries rather than authentic user behavior. We replaced the 120-prompt synthetic dataset with 300 real human prompts (50/category) sourced from production-system corpora on HuggingFace. This is a meaningful methodological shift: category-discriminative features identified on synthetic prompts may not generalize to real-world distributions. Using real prompts grounds our analysis in actual user behavior.

Old vs New Datasets

OLD (Milestone 1)	NEW (Milestone 2)
120 questions	**300 questions**
20 per category	50 per category
Made up by us	Real humans typed them
Short, neat	Diverse, messy
Reflects OUR ideas	Authentic behavior
(possibly biased!)	**(unbiased!)**

2.5x more questions, infinitely more authentic.

Chapter 8

Two Big Question Books

Where We Got Our 300 Questions

If I were 5...

We didn't just go ask people on the street. We went to TWO HUGE collections of questions that had already been asked online. Think of them like giant libraries of questions! Library #1 (Search Arena) is a SMART library where each book has a label saying "this is a math question" or "this is a story question." Library #2 (WildChat) is HUGE -- 4,800,000 questions! -- but a bit messier. We took some from each and mixed them together.

The real deal:

Source 1: Search Arena 24k (lmarena-ai/search-arena-24k) -- 24,069 conversations with expert-labeled primary_intent (Cohen's kappa = 0.812 inter-annotator agreement). Strong label reliability. Used for: factual, reasoning, creative.
Source 2: WildChat 4.8M (how2everything/WildChat-4.8M) -- 2.75M user-LLM messages, ~27 GB. We used HuggingFace's streaming API to scan ~2.5M rows on the fly without downloading everything. Used for: code, math (via keyword sub-classification), emotional.

Our Two Libraries

Library 1: Search Arena

Size	24,069 chats
Special	Each chat has a LABEL
Labels	"factual", "creative", etc.
Quality	Experts agree 81% of time
Best for	facts, reasoning, creative

Library 2: WildChat

Size	4,800,000 chats!
Special	Real conversations
Labels	Topic tags (code, etc.)
Trick	We "streamed" it (took peeks)
Best for	code, math, emotions

Chapter 9

Sorting Good Questions from Bad

Cleaning Up the Messy Stuff

If I were 5...

Real questions on the internet are MESSY. Some are too short ("hi"), some are way too long (a whole essay), some are URLs, some are gibberish, some say the same word over and over ("please please please please please..."). We needed to throw out the junk. So we made a 5-step sorting machine: each question has to PASS all 5 tests to be kept. It's like checking if a piece of fruit is good before putting it in your basket!

The real deal:

Multi-stage quality filter applied to every candidate prompt: (1) length bounds 3-200 words; (2) alpha ratio > 0.5 -- rejects URL/symbol-heavy text; (3) type-token ratio > 0.3 -- rejects repetitive spam; (4) GPT-2 token limit <= 512 to fit single forward pass without truncating last-token activations; (5) exact-match deduplication across both sources within each category. Only post-filter candidates are eligible for the final 50-per-category random sample.

The 5-Gate Quality Filter

1	**Length**	`3-200 words`	*Not too short, not too long*
2	**Letters**	`alpha > 50%`	*Real words, not symbols/URLs*
3	**Variety**	`TTR > 0.3`	*Not just "please please please"*
4	**Token Fit**	`<= 512 tokens`	*Fits in robot's brain*
5	**Unique**	`no duplicates`	*Each question is fresh*

Pass ALL 5 gates -> question is kept. Fail any -> rejected.

Chapter 10

Looking at the Questions

What We Found Before Even Asking the Robot

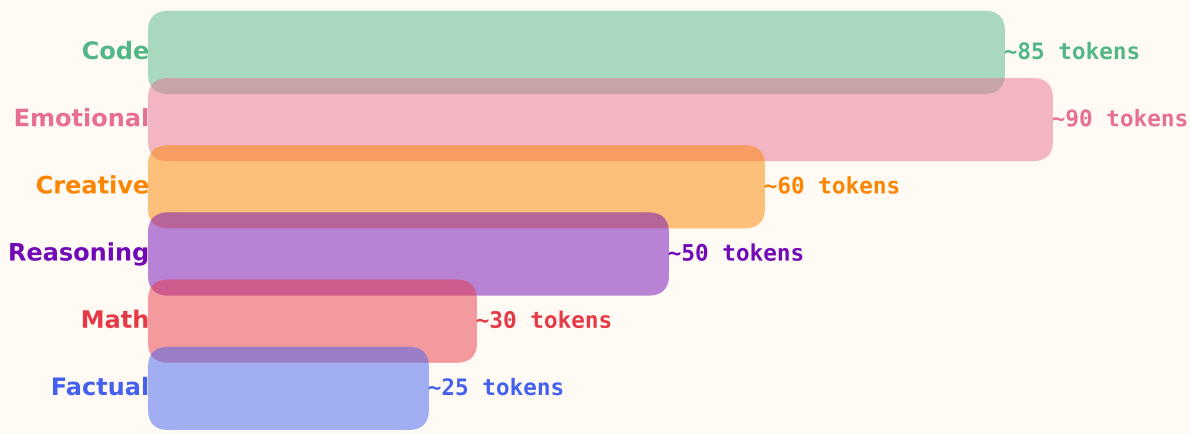
If I were 5...

Before showing the questions to the robot, we LOOKED at them carefully. And surprise! Different types of questions LOOK different even before any AI sees them. CODE questions are LONG (lots of typing). FACT questions are SHORT (just one specific question). EMOTIONAL questions are also long (people pour their feelings out). Math is short and punchy. So even the SHAPES of questions tell us something about what they are!

The real deal:

Pre-modeling exploratory data analysis revealed strong category-level distributional differences. Code prompts averaged ~85 GPT-2 tokens (longest). Factual prompts averaged ~25 tokens (most concise). Emotional prompts skewed long with high variance. Source composition is highly category-dependent: code is ~100% WildChat (computer_programming topic), factual is ~94% Search Arena. The new dataset has higher type-token ratio and longer length tail than the original synthetic 120-prompt set -- validating the M1->M2 dataset shift.

How Long Each Type of Question Is (avg tokens)



Why the big differences?

Code has function definitions and examples (long). Emotions have stories (long).
Math/factual questions are precise -- you ask one specific thing (short).

Chapter 11

Feature Extraction

Watching the Lights

If I were 5...

Now we watch carefully! We send each question into the robot's brain and put on special glasses (the SAE) that let us see which drawers light up. It's like an X-ray for robots! For each question, we check 3 floors (floor 2, floor 6, and floor 10) and write down every drawer that glows. Most drawers stay dark -- only a few dozen light up per question.

The real deal:

Extraction pipeline: (1) Tokenize prompt. (2) Forward pass with transformer_lens caching at hook_resid_pre for layers [2,6,10]. (3) Take last-token residual vector (shape [768]). (4) SAE.encode(resid) --> sparse feature vector (shape [24,576]). (5) Store as numpy array. The 'last token' is key: in causal attention, it aggregates information from all previous tokens, making it the richest representation of the full prompt.

How We Scan the Robot's Brain

1. Question goes in > **2. Words become numbers (tokens)** > **3. Travel through 12 floors** > **4. SAE sorts the jumble** > **5. Read which drawers glow!**

Why the LAST token?

The > cat > sat > on > the > mat > **.** **<-- This one!**

The last token has heard everything before it (causal attention).

It's like the student who sat through the entire lecture -- they know the most!

`Output: 300 prompts x 3 layers = 900 brain scans, each with 24,576 features`

Chapter 12

Sparsity

How Many Lights Turn On?

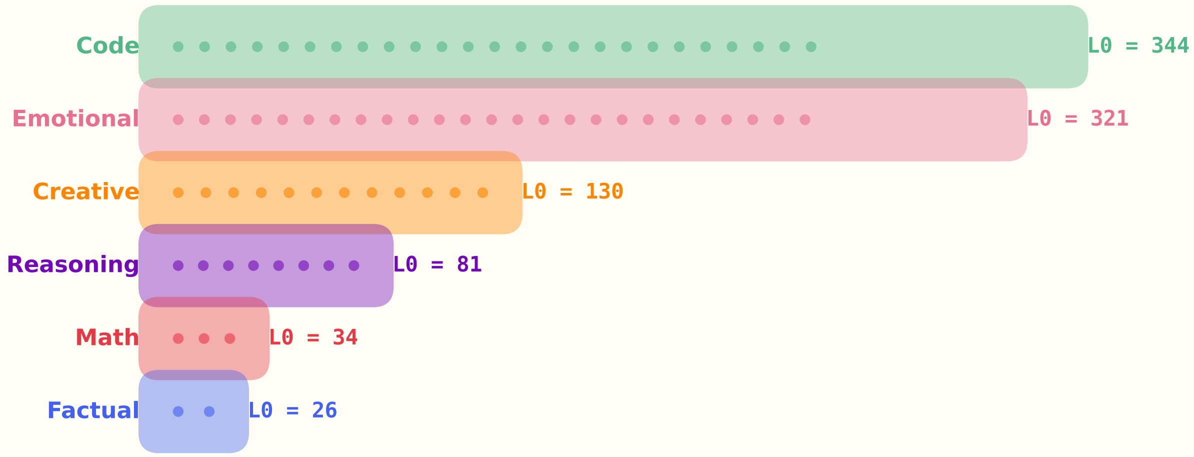
If I were 5...

Here's a surprising thing: out of 24,576 drawers, only about 50 to 300 actually light up for any question! That's like having a HUGE Christmas tree with 24,576 light bulbs, but only a handful turn on at a time. AND -- different types of questions turn on different AMOUNTS of lights! Coding questions turn on LOTS of lights (344!), like a big colorful party. Fact questions only turn on a few (26), like a quiet reading lamp. We call this L0 -- it just counts how many lights are on.

The real deal:

L0 (the number of nonzero SAE features) varies dramatically by category. At layer 2: code L0~344, emotional~321, creative~130, reasoning~81, math~34, factual~26. At layer 10, all categories converge: ~160-205. This reflects the representational complexity of each task -- code and emotional prompts require activating diverse features, while factual and math queries activate few but precise features.

How Many Lights Turn On? (Layer 2)



What does this mean?

Code and emotional prompts need MANY ideas at once (syntax + logic, or empathy + context). Factual/math needs just a FEW precise tools (recall + numbers). Broader tasks = more lights!

Chapter 13

Similarity & Clustering

Which Questions Look Alike Inside?

If I were 5...

If you look at the light patterns for two math questions, they'd look very SIMILAR -- almost the same drawers light up! But if you compare a math question to a poem question, the patterns look VERY different. It's like how two cats look similar to each other, but very different from a spaceship. We measured this by asking: "How much overlap is there between the lights for each type of question?" This is called cosine similarity.

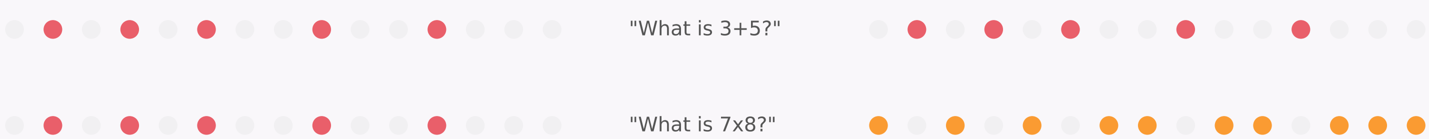
The real deal:

We compute the mean feature activation vector for each category and measure pairwise cosine similarity. At layer 2, creative prompts show the lowest similarity to all other categories, while math/reasoning are moderately similar. By layer 10, categories converge (higher similarity), suggesting the model transitions from input-type-specific representations to a more uniform format for next-token prediction. We also use UMAP to project the 24,576-dim feature space into 2D for visual clustering.

Do Similar Questions Have Similar Brain Patterns?

Math vs. Math — *(very similar patterns!)*

Math vs. Creative — *(very different patterns!)*



Same drawers light up!

Different drawers light up!

Cosine similarity measures how much two light patterns overlap.

High similarity = same type of thinking. Low similarity = totally different mental mode.

Chapter 14

Separability

Can We Tell Them Apart?

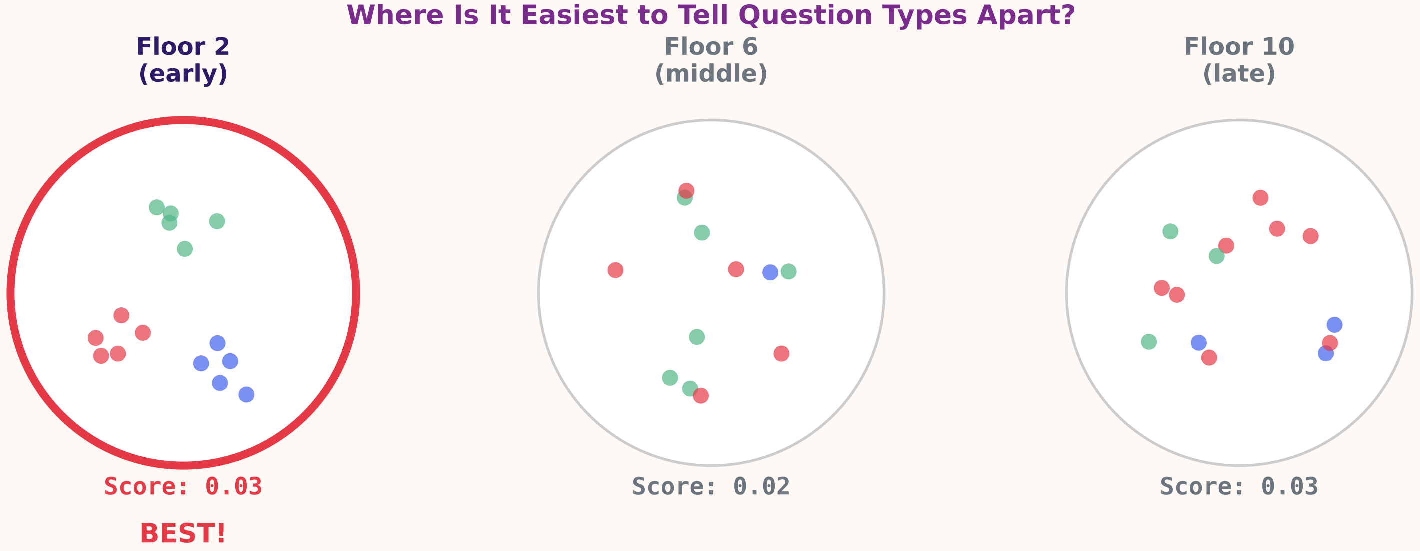
If I were 5...

If you showed me two crayon drawings -- one by a kid who loves red and one by a kid who loves blue -- I could easily tell whose is whose! But if both kids use ALL the colors, it's harder to tell. Separability asks: "At which floor of the building is it EASIEST to tell the question types apart?" The answer surprised us: floor 2 (the BOTTOM)! The robot figures out "this is a math question" very early, then mixes everything together as it goes up.

The real deal:

Fisher's discriminant ratio = between-class variance / within-class variance. A higher ratio means features better separate categories. Results: Layer 2 = 0.0287 (best), Layer 6 = 0.0235, Layer 10 = 0.0259. Layer 2's advantage suggests rapid input categorization in early layers, followed by convergence. Interestingly, layer 10 recovers slightly over layer 6, hinting at a U-shaped pattern as late layers prepare distinct output distributions.

Where Is It Easiest to Tell Question Types Apart?



Chapter 15

Activation Patching

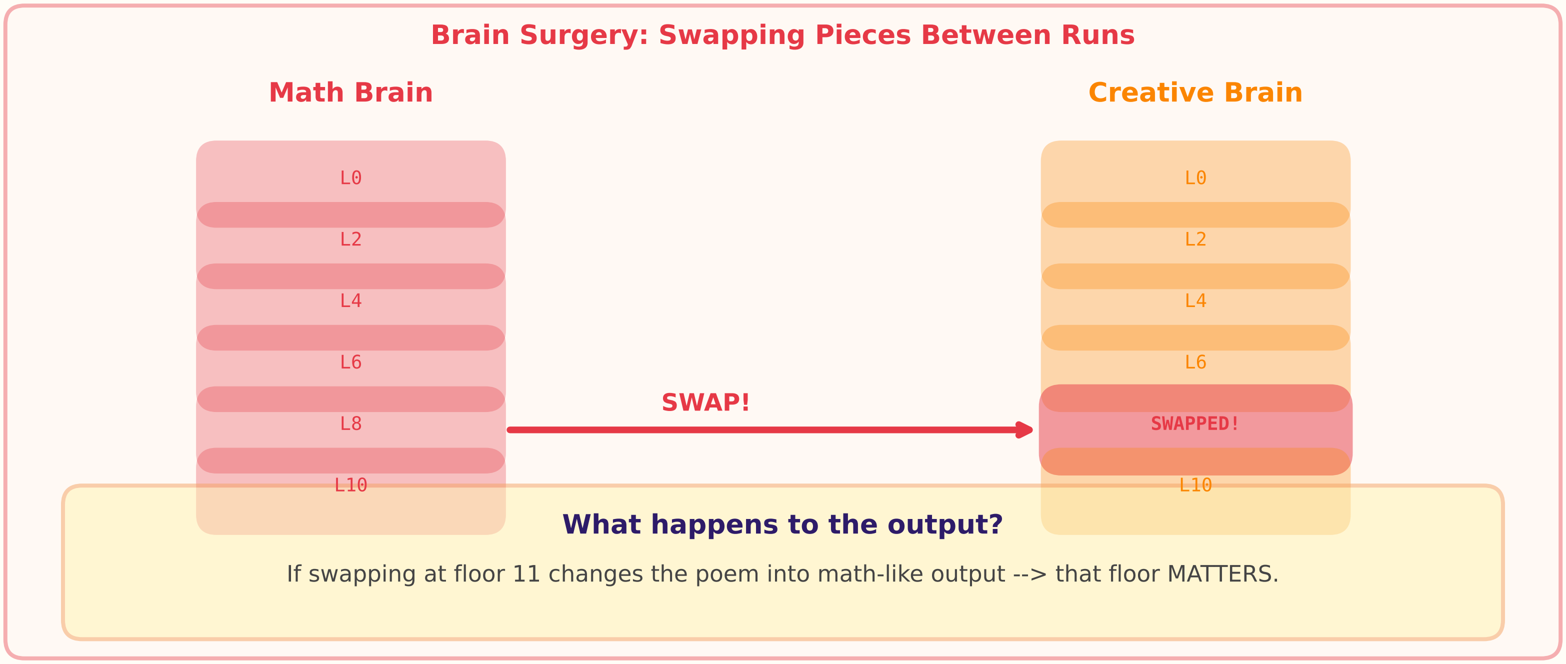
Brain Surgery on the Robot

If I were 5...

This is the coolest part! We do BRAIN SURGERY on the robot (don't worry, it doesn't hurt). We ask the robot two questions at the same time: a math question and a poetry question. Then we SWAP a piece of the math brain into the poetry brain at different floors and see what happens to the answer! If swapping at floor 11 makes the poem turn into math, we know floor 11 is where the "what to say" decision lives. It's like swapping ingredients in a recipe to find which one makes it spicy!

The real deal:

Activation patching: run source (math) and target (creative) prompts, cache all layer activations. Replace the target's residual stream at layer L with the source's. Measure KL divergence between patched output and clean target output. Lower KL = layer L carries more causal information about what the model will output. We sweep all 12 layers. Result: monotonically decreasing KL from layer 0 to 11. Layer 11 (KL~0) is most causally important.



Chapter 15 (continued)

Surgical vs. Full Brain Swap

The distributed nature of knowledge

If I were 5...

We tried two kinds of surgery: (1) swapping the ENTIRE floor (all 768 slots), and (2) swapping just the 10 most important drawers. Guess what? Swapping the whole floor worked great -- the poem started to look like math! But swapping just 10 drawers barely did anything. This means the "math knowledge" isn't hiding in just a few special drawers -- it's SPREAD OUT across hundreds of drawers, like butter on toast!

The real deal:

Full residual patch at layer 6: KL = 0.64 (strong shift). Surgical patch of top-10 discriminative feature directions: KL = 3.48 (baseline = 3.61, so only a 0.13 reduction). The top-10 features account for ~4.5% of the total causal effect. Conclusion: math-relevant information is distributed across many features in superposition, not concentrated in a few 'math neurons'. This is consistent with the superposition hypothesis.

Full Swap

(entire floor)

All 768 dimensions swapped

KL = 0.64 (BIG change!)

Surgical Swap

(top 10 features only)

Only 10 of 24,576 features

KL = 3.48 (tiny change...)

The Big Lesson:

Math knowledge isn't in 10 special drawers. It's SPREAD across hundreds.

Like butter on toast -- you can't pick out just the butter.

(This is what 'superposition' looks like in practice.)

Chapter 16

Feature Steering

Pushing the Robot's Thoughts

If I were 5...

What if you could push the robot's thoughts in a direction? Like, the robot is thinking about poetry, but you sneak in and turn the "math" dial up to 20x! Will the robot start writing math instead of poems? We tried this! We amplified the 10 strongest math drawers by 20 times while the robot was thinking about a neutral prompt. The output changed a bit -- but didn't become clearly math-like. The math thoughts are too spread out to push with just 10 drawers.

The real deal:

Feature steering: during generation, we add 20x the top-10 math feature directions to the residual stream at layer 6. This modifies the model's internal state to amplify math-related concepts. Result: output was altered but not clearly math-like, confirming the distributed nature finding from patching. More aggressive steering (100+ features, multi-layer) would be needed for stronger effects.

Turning the Math Dial While Thinking About Poetry

Before Steering:

Input: "Write a poem"

Output: beautiful poem

Math dial 20x! →

After Steering:

Input: "Write a poem"

Output: slightly weird poem

Why didn't it become math?

Because 10 features out of 24,576 is like turning up 10 pixels on a big screen -- the picture barely changes. You'd need hundreds of features to really shift the output.

Chapter 17

Zooming In on the Stars

Picking the Most Important Lights to Study Closely

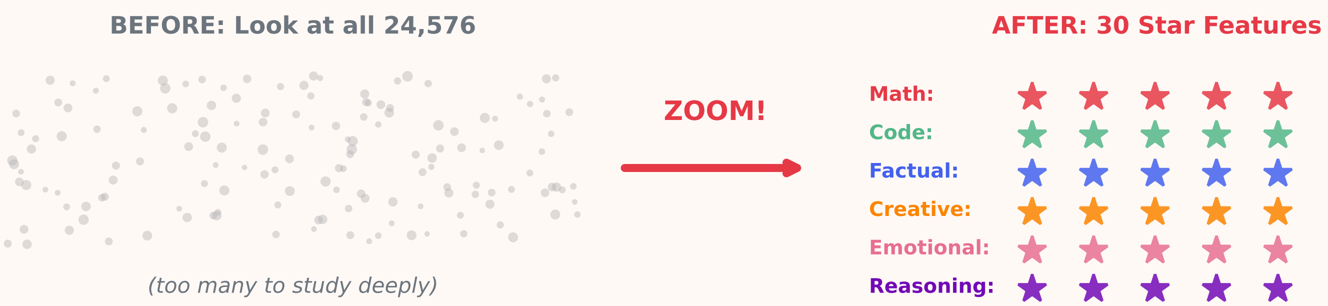
If I were 5...

Out of 24,576 light bulbs, we don't have time to study EVERY one. So our teacher said: "Pick the BRIGHTEST stars -- the top 5 most important light bulbs for each type of question -- and study THOSE in detail!" It's like instead of looking at every single tree in a forest, you pick out the tallest, weirdest-shaped trees and learn everything about them. Which trees they are. Why they grew that way. What animals live in them. We'll do that with the most-discriminative SAE features!

The real deal:

Per professor's most recent feedback, the highest-priority next step is narrowing analytical focus from broad surveys to deep dives on a small set of carefully-selected features. For each top-5 discriminative feature per category (e.g., #22431 reasoning, #1881 code, #7484 factual), the planned analysis is: (1) retrieve max-activating examples from Neuronpedia, (2) write feature interpretations grounded in those examples, (3) run feature-level steering experiments, (4) inspect attention heads writing to those feature directions to build circuit-level explanations. This is the path from 'features exist' to 'we understand them.'

From 24,576 -> Top 5 per Category = 30 Stars to Study



For each of the 30 star features, we'll find out:

* What it MEANS (look up examples on Neuronpedia)
* WHEN it fires (which questions trigger it)
* WHY it matters (does turning it up change the answer?)
* WHO talks to it (which attention heads write to it)

Chapter 18

What We Discovered

The Big Answers

YES, different questions light up different drawers!

Each prompt category has a distinct feature activation signature. The model genuinely processes math differently from poetry at the feature level.

The robot figures out the question type EARLY!

Layer 2 (floor 2) has the best separability (Fisher = 0.029). The model categorizes inputs quickly, then blends for generation.

Code and emotional thinking are BROAD, factual thinking is NARROW!

Code L0 = 344 features vs. factual L0 = 26 at layer 2. Code and emotional prompts activate 13x more features -- broader conceptual demands.

Knowledge is SPREAD like butter, not in one jar!

Surgical patching of top-10 features captured only 4.5% of the effect. Information is distributed across hundreds of features in superposition.

Reasoning has the strongest single feature!

Feature #22431 has selectivity score 2.11 for reasoning prompts -- the single most category-specific feature. It's also shared with emotional prompts, suggesting a 'complex query' detector.

Later floors make the final decisions!

Layer 11 is most causally important (KL -> 0 when patched). The model makes output decisions in the final layers.

Quick Reference

The Friendly Glossary

Every big word, explained simply

Term	Friendly Version	Expert Version
LLM	*A word-guessing robot*	Large Language Model -- predicts next tokens autoregressively
Layer	*A floor in the building*	One transformer block (attention + MLP + residual connection)
Residual Stream	*The hallway between floors*	768-dim vector carrying accumulated information
Superposition	*Too many toys in one box*	Encoding more features than dimensions via near-orthogonality
SAE	*The magic sorting machine*	Sparse Autoencoder -- decomposes 768 dims into 24,576 features
Feature	*One labeled drawer*	A monosemantic direction in activation space
L0 (Sparsity)	*How many lights are on*	Count of nonzero SAE feature activations
Cosine Similarity	*Pattern overlap*	Dot product of normalized vectors; measures direction alignment
Fisher Ratio	*How easy to tell apart*	Between-class / within-class variance
Activation Patching	*Brain surgery*	Swap activations between runs to measure causal importance
KL Divergence	*How different the answer got*	Information-theoretic measure of distribution difference
Feature Steering	*Pushing the dial*	Amplifying feature directions to steer model output

Now you know what's happening

inside the AI's brain!

Whether you're five or fifty,

the ideas are the same --

just the words get bigger.

Model: GPT2 | SAE: gpt2-small-res-jb | 24,576 features

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